

SJAA Ephemeris Bulletin

✱ August 2025

Dark Sky Star Party

Astronomical outreach is a core part of SJAA's mission. Yes, that's right, we have a *mission*: "to provide experience with and education about astronomy and related sciences to members and the public." What better way to achieve this mission than a public star party?

SJAA hosts a variety of Star Parties in all shapes and sizes. We try to meet people where they are with our In-Town Star Party (ITSP) at Hough Park every month, and with Solar Sundays. We have dedicated volunteers who coordinate and implement star parties for students in schools across the Bay Area. In the summertime, SJAA coordinates with Pinnacles National Park to put on star parties for the public under the darkest skies we have regular access to. And just this month, we hosted two nights of public star parties in Yosemite National Park at Glacier Point, under the watchful gaze of Half Dome.

Through these activities, we aim to expand public awareness about astronomy. We are fortunate to mingle with people from our community, state, country, and all over the world, and to inspire passion for the cosmos. Personal observation of the night sky is enjoyable for us amateur astronomers, but outreach is transcendent. Keep looking up... and while you're at it, share the view!

Carl Svensson
Vice-President
Ephemeris Editor

Target of the Month: M20

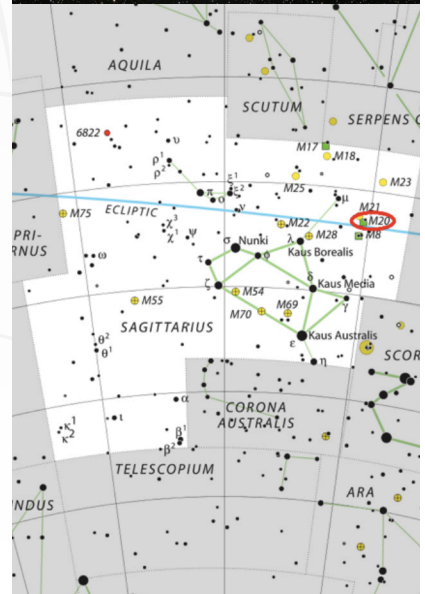
This month's target is Messier 20, the Trifid Nebula in the constellation Sagittarius. It is located approximately 5200 light years away and was discovered by Charles Messier on June 5, 1764.

Its name means 'three-lobed'. The object is an unusual combination of multiple types of objects. It is an open star cluster, an emission nebula characterized by the "red" color resulting from ionized hydrogen gas, a reflection nebula characterized by the "blue" color resulting from starlight that is reflected by gas and dust, and a dark nebula characterized by black bands of dense gas that light cannot penetrate.

Some of the stars found within the Trifid Nebula are only a few hundred thousand years old, making the open cluster in M20 one of the youngest open clusters known. Some of the extremely young and massive stars that formed in the region may have already ended their lives in supernova explosions which may be responsible for the appearance of the dust lanes in M20.

The Trifid Nebula can be found about 2 degrees to the northwest of the larger and brighter Lagoon Nebula (Messier 8) and appears in the same field of view as the open cluster Messier 21. Messier 20 can be located using the Teapot asterism in Sagittarius. It lies a few degrees to the west of Lambda Sagittarii, the star at the top of the Teapot. From a dark sky the Trifid Nebula and the Lagoon Nebula can be detected by the naked eye as fuzzy patches embedded in the Milky Way. With some imagination, the position of the nebulae, above the "spout" of the Teapot, give the appearance of rising steam.

Magnitude	6.3
Size	42 ly
Object Type	Nebula + Open Cluster
Constellation	Sagittarius
Distance	5200 ly
Age	300,000 years



Top: M20 on July 18th at Hough Park by Joe Fragola. 35 Minutes with Celestron Origin. Bottom: M20 Location by IAU and Sky & Telescope

Announcements

Grant Ranch Star Party There will be a public Star Party at the Grant Ranch County Park on August 16, 2025. It has been more than 5 years since the last public Star Party at the park! See: Star Party | Santa Clara County Parks | County of Santa Clara for more information. The Star Party will be held in the Park adjacent to the Stockman Picnic area on a paved hiking path leading from the Stockman parking lot. The Park entrance station located on Mt Hamilton Road just past Quimby Road (right turn into the Park). Grant Ranch County Park has decently dark skies (most moonless nights the stars in the Little Dipper are visible).

Note that this is not a SJAA event, but the Park staff is

asking for telescope volunteers with public star party experience. There are 10 volunteer slots available. If you are interested in volunteering, please contact Larry Zurbrich at president@sjaa.net. Note also that this event will be held on the same evening as the SJAA Binocular Stargazing and Starry Night.

We Always Need More Volunteers There are many volunteer opportunities in SJAA! While we love having seasoned experts in our volunteer ranks, we happily welcome enthusiastic beginners, too. Opportunities range from operating telescopes, to authoring content for publication, to helping with event logistics. Please reach out to us at volunteer@sjaa.net to get started!



SJAA August 2025

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
					1 Venus 2° S of M35 Helen Sawyer Hogg's Birthday (1905)  Maria Mitchell's Birthday (1818) 56%	2  65%
3 Solar Sunday, Fix-It 2:00PM Spica 0.8° N of the Moon  Antares 0.6° N of Moon 74%	4  83%	5  90%	6 Saturn 1.1° S of Neptune  95%	7 Pluto 0.1° S of Moon  95%	8 Astro 101: Binoculars 7:00PM Full Moon (Sturgeon Moon)  100%	9 Board Meeting 6:00PM Guest Speaker 7:30PM  100%
10  99%	11  95%	12 Peak of Perseid Meteor Shower Astro 101: Beginner's Forum 7:00PM  Venus 0.9° S of Jupiter 88%	13  80%	14  69%	15  58%	16 Binocular Stargazing 8:30PM Starry Nights 9:00PM  Moon 0.9° N of the Pleiades 46%
17  34%	18  24%	19 Imaging SIG 7:00PM Jupiter 5° S of Moon  Milton Humason's Birthday (1891) 14%	20 Venus 5° S of Moon  7%	21 Moon 2° N of M44 Mercury 4° S of Moon  3%	22 Astro 101 7:45PM In-Town Star Party 9:00PM  New Moon 0%	23 Mendoza Ranch Member Observing 8:45 PM  0%
24  Andrew Fraknoi's Birthday (1948) 3%	25  7%	26 Mars 3° N of Moon  13%	27 Spica 1.2° N of the Moon  21%	28  29%	29  39%	30  First Quarter Moon 48%
31 Antares 0.7° N of Moon  58%			Times and dates are subject to change. Please check sjaa.net and www.meetup.com/sj-astronomy for the latest updates.			 Meteor Shower  Dark Sky Window